RMRRT: Riemannian Barrier Metric RRT for Inequality-Aware Steering on Equality Manifolds

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***Abstract*—This paper presents a novel motion planning framework for high-dimensional robotic systems subject to simultaneous equality and inequality constraints. Conventional sampling-based planners often exhibit inefficient exploration because they treat constraints separately: equality constraints via projection and inequality constraints via binary collision checking. To address this issue, we propose Riemannian Barrier Metric RRT (RMRRT), which unifies both constraints into a single geometric formulation. RMRRT constructs a Riemannian barrier metric on the tangent space of the equality manifold by integrating the null-space projection of the constraint Jacobian with the Hessian of a logarithmic barrier function. This metric guides the tree expansion by selecting steering directions that minimize the induced Riemannian norm, thereby naturally avoiding inequality constraints while strictly adhering to the equality manifold. Experimental results show that RMRRT significantly reduces collision-induced sample rejection and achieves superior planning efficiency and success rates compared to state-of-the-art constrained planners. Experiment videos and source code are available at: https://rcilab.github.io/rmrrt**

***Index Terms*—Constrained motion planning, Sampling-based planning, Riemannian manifold, Inequality constraints, Rapidly-exploring random tree.**

I. Introduction

SAMPLING-based motion planning becomes particularly challenging when robotic systems are subject to simultaneous equality and inequality constraints [1]–[3]. Equality constraints commonly arise from closed-chain kinematics [4]–[6], grasp maintenance [7]–[9], or cooperative multi-arm manipulation [10], [11], restricting feasible configurations to an implicit manifold embedded in the ambient configuration space [12]–[14]. At the same time, inequality constraints—such as obstacle avoidance [15], [16], self-collision [17]–[19], and joint limits [2], [20]—further restrict admissible regions through state-dependent boundaries. Under these conditions, sampling-based planners must efficiently explore admissible regions that occupy only a small fraction of the configuration space, making exploration efficiency and steering quality critical to planning performance.

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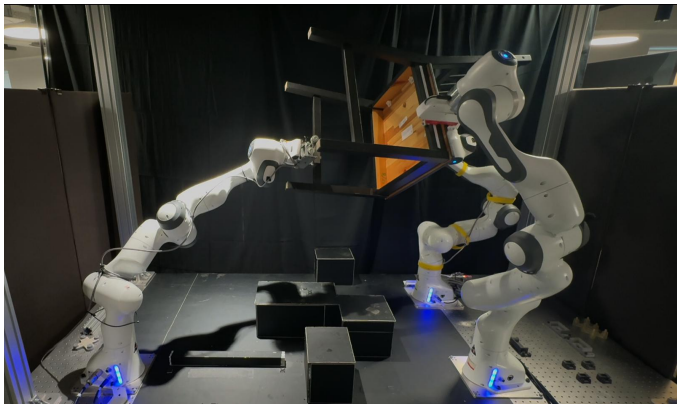

Fig. 1: ~~Demonstration of RMRRT in a cooperative transport scenario.~~ Real-world demonstration of RMRRT on a triple-arm Panda system cooperatively transporting a chair. The task involves simultaneous equality constraints from closed-chain grasp maintenance and inequality constraints from obstacle and self-collision avoidance.

In conventional sampling-based planners, equality constraints are typically enforced through projection onto the constraint manifold, while inequality constraints are handled separately via binary feasibility checking such as collision testing [12]–[14], [21]. This decoupled treatment can lead to inefficient sample rejection and unstable steering behavior, particularly in highly constrained environments where admissible regions are narrow or fragmented. Consequently, integrating equality and inequality constraints—whose geometric characteristics are inherently distinct—into a unified exploration framework remains a central challenge in sampling-based motion planning. In this work, we address this challenge by proposing a unified geometric formulation that directly encodes inequality-induced boundary geometry into the exploration process on equality-constrained manifolds.

A. Related Works

Sampling-based robot motion planning under constraints has been studied from multiple perspectives, largely distinguished by how equality and inequality constraints are represented and integrated into the exploration process. Prior work can be broadly categorized into three groups: (i) manifold-based approaches that explore directly on equality-constrained manifolds, (ii) metric-based approaches that encode inequality-induced geometry into distance or metric structures, and (iii) data-driven approaches that accelerate sampling and steering using learned or preconstructed priors.

1) Manifold-Based Motion Planning: Manifold-based motion planning explores directly on implicitly defined equality-

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constrained manifolds, aiming to maintain equality feasibility throughout the planning process [5], [12]–[14]. Atlas-based planners approximate the manifold using a collection of local charts and perform global exploration by transitioning across neighboring charts during tree expansion, thereby improving numerical robustness even in highly curved regions [13], [22]. Tangent-space approaches, in turn, generate steering directions in the Jacobian-defined local tangent space and apply manifold-consistent correction (e.g., local projection) to produce equality-consistent extensions [12], [14]. However, in many implementations, tangent-space information is used primarily to ensure equality-feasible extensions, whereas nearest-node selection and steering criteria are not consistently informed by inequality-induced local geometry. Consequently, inequality constraints often influence exploration only through post-hoc rejection via external binary validity checks such as collision testing [23], which can lead to many unproductive extension attempts in narrow passages or heavily cluttered environments and substantially degrade exploration efficiency [24], [25]. ~~In contrast, our approach maintains exploration on the equality manifold while using a metric that reflects inequality-induced geometry to inform both nearest-node selection and steering, thereby reducing collision-driven rejection at its source.~~

2) Metric-Based Motion Planning: Metric-based approaches incorporate inequality constraints into costs, distance measures, or position-dependent metrics to discourage motions near constraint boundaries and bias exploration toward admissible regions. For example, artificial potential fields and continuous cost-based formulations encode obstacle proximity in a smooth and differentiable manner [26], [27], and more recent Riemannian formulations construct non-Euclidean, state-dependent metrics using curvature or Hessian information associated with inequality constraints [28], [29]. Relatedly, reactive motion policy frameworks such as RMPflow [30] compose task-space Riemannian motion policies to generate real-time local control commands under multiple objectives. While effective for reactive behaviors, these methods are not designed to perform sampling-based global exploration on implicit equality-constrained manifolds, and typically require additional mechanisms to strictly enforce closed-chain or grasp-maintenance constraints. Moreover, many metric-based approaches are formulated in unconstrained configuration spaces or incorporate equality constraints only through soft penalties; as a result, additional projection or correction steps are commonly required to ensure consistent steering and tree expansion on equality-constrained manifolds. ~~Our approach differs in that the Riemannian barrier metric is constructed directly on the tangent space of the equality manifold, thereby jointly encoding both constraint types within a single geometric structure without requiring soft penalties or additional correction mechanisms.~~

3) Data-driven Motion Planning: Data-driven approaches accelerate constrained planning by capturing the structure of the feasible space through data or preconstructed priors, thereby biasing sampling distributions and expansion decisions toward favorable regions. In particular, prior work has explored learning low-dimensional latent representations of equality-constrained manifolds [6], [31] to reduce the effective search dimension, as well as using learned samplers or policies to guide exploration. Complementary to learning-based priors, offline structure-based methods such as precomputed graph (roadmap) approaches [5], [32] improve online performance by leveraging roadmaps constructed in advance from constraint-satisfying states to reduce the online search burden. However, the effectiveness of these data- or structure-driven techniques can be sensitive to data coverage or to the density and suitability of the precomputed structures; moreover, maintaining comparable performance under new environments or changed constraint conditions is difficult to guarantee, and extending the applicability range may require additional data collection or structure updates. ~~Our approach does not rely on learned priors or precomputed structures, yet achieves competitive performance through a purely geometric formulation. Furthermore, the proposed metric can serve as a complementary module within data-driven planners to enhance their geometric consistency during exploration.~~

B. Overview of Our Approach

As discussed above, manifold-based planners maintain equality feasibility but handle inequality constraints only through post-hoc rejection, metric-based approaches encode inequality geometry but lack strict equality-manifold consistency, and data-driven methods rely on environment-specific priors that may not generalize. To address these limitations, we propose Riemannian Barrier Metric RRT (RMRRT), a sampling-based planner that enables inequality-aware steering while preserving feasibility on an equality-constrained manifold. The central idea is to endow equality-consistent motions with a Riemannian barrier metric that explicitly encodes the local geometry induced by inequality constraints (e.g., obstacle avoidance, self-collision, and joint limits). Concretely, we construct a state-dependent metric by combining the null-space projection associated with the equality-constraint Jacobian and the Hessian of a logarithmic barrier function defined over inequality constraints, yielding a unified geometric structure for exploration on the manifold.

Based on this metric, RMRRT selects expansion directions by approximately minimizing the induced Riemannian norm within the equality-consistent tangent space, thereby naturally discouraging motions toward inequality boundaries already at the steering stage. The same metric also provides a geometry-consistent notion of local connectability between states, and is therefore leveraged in nearest-node selection to prioritize nodes that admit productive manifold-consistent progress. Consequently, inequality constraints are no longer treated as a post-hoc validity filter; instead, tree growth is directly guided along admissible regions, reducing collision-driven rejections and improving exploration efficiency in highly constrained environments. In this work, we instantiate the proposed metric within a bidirectional RRT-Connect framework [33], resulting in a planner that remains compatible with standard sampling-based planning pipelines while explicitly accounting for inequality constraints during both selection and steering.

The main contributions of this paper are summarized as follows:

- We propose a Riemannian barrier metric that unifies equality-manifold structure and inequality-induced boundary geometry within a single formulation.
- Building on this metric, we develop a bidirectional RRT-based planner, RMRRT, that performs inequality-aware exploration while maintaining feasibility on equality-constrained manifolds.
- We incorporate the proposed geometry consistently into both steering and nearest-node selection, guiding expansion away from infeasible regions and reducing validity-driven sample rejection during tree growth.
- We validate the proposed method through extensive simulation and real-world experiments on constrained manipulation tasks, including multi-arm and humanoid systems, demonstrating improved planning efficiency and reliability compared to existing constrained planners.

The remainder of this paper is organized as follows. Section II introduces the necessary preliminaries, including equality-constrained manifolds and barrier-based metrics. Section III presents the proposed RMRRT methodology in detail. Section IV reports experimental results on a variety of constrained manipulation scenarios. Finally, Section V concludes the paper and discusses directions for future work.

II. Preliminaries

This section summarizes the notation and geometric tools required to develop the proposed method. Specifically, we formalize the motion planning problem under equality and inequality constraints, review the geometric structure of equality-constrained manifolds and their associated tangent spaces, and summarize barrier-based representations for inequality constraints, which will be used to construct the proposed Riemannian barrier metric in Section III.

A. Problem Formulation

We consider motion planning problems for robotic systems operating in a continuous configuration space $\mathcal{Q} \subset \mathbb{R}^n$, where a configuration is denoted by $q \in \mathcal{Q}$. The system is subject to both equality and inequality constraints, which jointly define the feasible configuration space. Equality constraints are expressed as

$$g(q) = 0, \tag{1}$$

where $g : \mathbb{R}^n \to \mathbb{R}^m$ is a differentiable constraint function. These constraints restrict feasible configurations to an implicitly defined manifold embedded in the ambient configuration space.

In addition, inequality constraints are given by

$$h_i(q) > 0, \quad i = 1, \dots, k \tag{2}$$

where each $h_i : \mathbb{R}^n \to \mathbb{R}$ represents a state-dependent constraint such as obstacle avoidance [15], [16], self-collision avoidance [17]–[19], or joint limits [2], [20]. Together, the inequality constraints define admissible regions within the equality-constrained manifold. The overall feasible configuration set is therefore defined as

$$\mathcal{F} = \{q \in \mathcal{Q} \mid g(q) = 0,\ h_i(q) > 0,\ \forall i\}. \tag{3}$$

We assume that valid start and goal configurations lie within $\mathcal{F}$, and focus on planning collision-free motions that remain feasible throughout the planning process. In sampling-based motion planning, this requires generating tree expansions that preserve equality constraints while avoiding violations of inequality constraints.

B. Equality-Constrained Manifolds and Tangent Spaces

Equality constraints of the form $g(q) = 0$ define an implicit manifold embedded in the ambient configuration space $\mathcal{Q}$. Specifically, the equality-constrained manifold is given by

$$\mathcal{M} = \{\, q \in \mathcal{Q} \mid g(q) = 0 \,\} \tag{4}$$

Let $J(q)$ denote the Jacobian of the constraint function,

$$J(q) = \frac{\partial g(q)}{\partial q}. \tag{5}$$

If $J(q)$ has full row rank m for all $q \in \mathcal{M}$, then $\mathcal{M}$ is a smooth manifold of dimension $n - m$[1].

At a configuration $q \in \mathcal{M}$, the tangent space of $\mathcal{M}$ is defined as

$$T_q\mathcal{M} = \{\, v \in \mathbb{R}^n \mid J(q)v = 0 \,\}. \tag{6}$$

This space characterizes the set of infinitesimal velocities that preserve the equality constraints to first order. In sampling-based motion planning, tangent directions are used to construct local expansions, and a retraction is applied when needed to keep the expanded state on the manifold $\mathcal{M}$. Moreover, when $J(q)$ has full row rank m, the matrix $J(q)J(q)^\top$ is invertible, which allows defining the orthogonal projector onto $T_q\mathcal{M}$ as

$$Q(q) = I - J(q)^\top \left(J(q)J(q)^\top\right)^{-1} J(q) \tag{7}$$

where I denotes the identity matrix. The projector $Q(q)$ satisfies $J(q)Q(q) = 0$ and maps arbitrary vectors in $\mathbb{R}^n$ onto the tangent space $T_q\mathcal{M}$. This tangent-space formulation provides a fundamental tool for constructing constraint-consistent steering and expansion operators, which will be exploited in the proposed approach.

C. Inequality Constraints and Log-Barrier Functions

As introduced in ~~Eq.~~ (2), the inequality constraints induce state-dependent boundaries that restrict admissible configurations within the equality-constrained manifold. In sampling-based planning, such constraints are typically handled via binary feasibility checks. To obtain a continuous representation of inequality constraints, we adopt a log-barrier formulation. Specifically, given a set of k inequality constraints $\{h_i(q)\}_{i=1}^k$, we define the logarithmic barrier function

$$\phi(q) = -\sum_{i=1}^{k} \log h_i(q). \tag{8}$$

This function grows rapidly as a configuration q approaches the boundary of the admissible region, thereby smoothly encoding proximity to inequality constraints. Moreover, the Hessian of the barrier function captures second-order information associated with the curvature of the constraint boundaries, providing a principled means to quantify how inequality constraints locally distort the geometry of the configuration space. ~~In constrained motion planning, barrier-based representations offer a smooth alternative to binary collision checking by continuously reflecting the influence of inequality constraints~~

[1]This condition corresponds to the standard regularity assumption of the implicit function theorem.

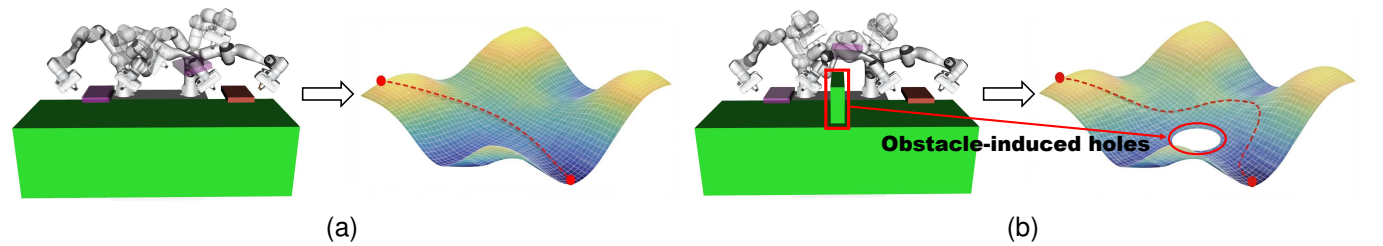

Fig. 2: Conceptual illustration of inequality-induced geometry on an equality-constrained manifold. (a) Equality-constrained manifold under the implicit assumption of full admissibility, as commonly treated in tangent-space and projection-based planners. (b) The same manifold after removing infeasible regions induced by inequality constraints, which form holes in the feasible subset. Although the equality constraint is unchanged, the admissible region exhibits nontrivial geometry that is not captured by conventional steering. RMRRT accounts for this geometry through a Riemannian metric defined on the tangent space.

~~within admissible regions.~~ This property makes log-barrier functions well-suited for defining geometry-aware structures that can be leveraged during planning, as will be described in ~~subsequent sections~~ Section III.

III. Riemannian Barrier Metric RRT

In this section, we present the proposed Riemannian Barrier Metric RRT (RMRRT). RMRRT is a sampling-based planning framework that unifies equality and inequality constraints through a single geometric formulation. Building on the equality-manifold structure, tangent-space representation, and log-barrier modeling introduced in Section II, RMRRT modifies the local expansion mechanism of RRT planners so that exploration explicitly reflects constraint-induced geometry. The remainder of this section describes the overall framework, the construction of the Riemannian barrier metric, and its integration into a bidirectional RRT-Connect planning algorithm.

A. Proposed Framework

The proposed RMRRT builds on a bidirectional RRT-Connect backbone and modifies the local expansion mechanism to account for constraint-induced geometry on the equality-constrained manifold $\mathcal{M}$. While retaining the canonical pipeline of random sampling, nearest-neighbor selection, and incremental tree extension, RMRRT makes these operations geometry-aware by using a metric-induced distance for neighbor queries and incorporating a Riemannian barrier metric during steering. Fig. 2a illustrates an idealized view in which no regions are carved out by inequality constraints, under which steering is commonly constructed in many tangent-space and projection-based approaches [5], [12], [13]. In contrast, Fig. 2b shows that inequality constraints create infeasible regions (holes) on the same manifold $\mathcal{M}$, making the feasible subset exhibit nontrivial structure. In such environments, tangent-space steering in $T_q\mathcal{M}$ alone repeatedly attempts expansions toward infeasible regions, leading to increased rejection rates and unstable exploration. To mitigate this issue, RMRRT discourages motions toward inequality boundaries via the Riemannian barrier metric, thereby biasing local expansions along feasible regions. The following subsections describe the metric construction and its integration into the planning pipeline.

Algorithm 1 BuildMetric(q): Riemannian barrier metric

Require: Configuration q
Ensure: Ambient metric $G(q)$, tangent projector $P(q)$, Riemannian metric $M(q)$

for $j = 1, \ldots, m$ **(proxy points) do**
  Compute proxy point $p_j(q)$ and signed distance $d_j(q)$
  Compute point Jacobian $J_j(q) = \partial p_j(q)/\partial q$
end for

$G(q) \leftarrow \lambda I + \sum_{j=1}^{m} w(d_j(q))\, J_j^\top(q) J_j(q)$ (App. A)

Compute equality Jacobian $J(q)$
$A \leftarrow J(q)\, G^{-1}(q)\, J^\top(q)$
$P(q) \leftarrow I - G^{-1}(q) J^\top(q)\, A^{-1}\, J(q)$ (App. B)
$M(q) \leftarrow P^\top(q)\, G(q)\, P(q)$

return $G(q), P(q), M(q)$

B. Riemannian Barrier Metric Construction

Leveraging the barrier-based formulation of inequality constraints introduced in Eq. (8), we construct a Riemannian metric that encodes how proximity to inequality boundaries reshapes the local geometry of the configuration space. In principle, the ambient metric is defined by accumulating the exact Hessians of the inequality barrier functions:

$$G(q) = I + \sum_{i=1}^{k} \alpha_i \nabla^2 \phi_i(q), \tag{9}$$

where ϕ_i denotes the log-barrier associated with the i-th inequality constraint and $\alpha_i > 0$ is a weighting factor. However, in cluttered manipulation scenes, $\phi_i(q)$ typically depends on signed distances and exhibits frequent closest-feature changes, making direct evaluation of $\nabla^2\phi_i(q)$ computationally expensive and potentially numerically fragile.

To enable efficient and stable metric updates during planning, we therefore adopt a Gauss–Newton (GN) Jacobian-based surrogate that preserves the essential effect of penalizing motions toward inequality boundaries while avoiding explicit second-order derivatives. Specifically, we introduce a set of geometric proxy points $p_j(q) \in \mathbb{R}^3$ on the robot and the manipulated object, together with their signed distances to the environment $d_j(q)$ and point Jacobians $J_j(q)$. The ambient

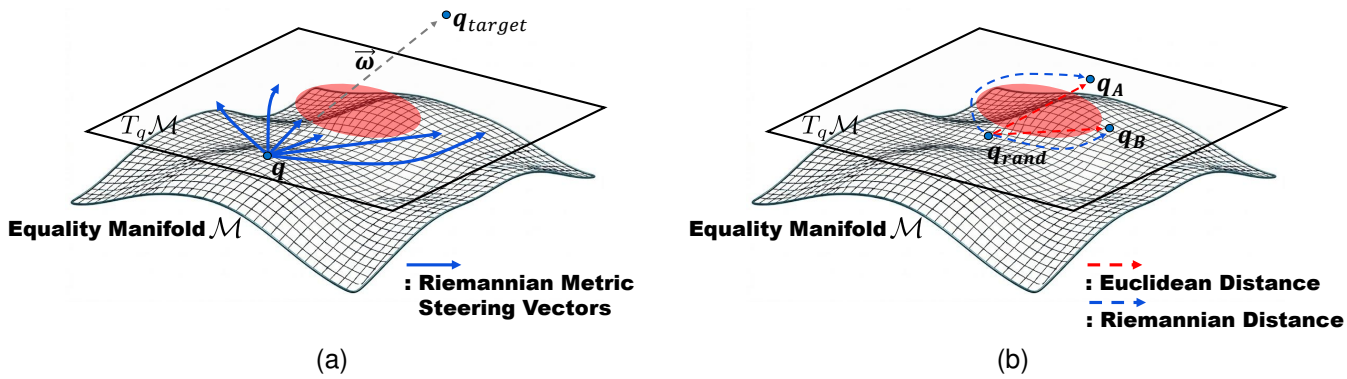

Fig. 3: Conceptual illustration of the Riemannian metric-based exploration mechanism in RMRRT. (a) Metric-induced tangent-space steering. Given a random target, the induced direction is restricted to the tangent space of the equality-constrained manifold at configuration q. Directions that move toward inequality boundaries incur higher cost under the Riemannian barrier metric, thereby biasing steering toward admissible regions. (b) Riemannian metric-based nearest-neighbor selection. Nodes that appear similar in Euclidean space can be separated under the Riemannian distance due to inequality-induced geometry. Consequently, RMRRT preferentially expands nodes that avoid high-cost regions near constraint boundaries, improving exploration stability and expansion efficiency.

metric is then constructed in a PSD outer-product form as

$$G(q) = \lambda I + \sum_{j=1}^{m} w(d_j(q))\; J_j(q)^\top J_j(q), \tag{10}$$

where $\lambda > 0$ is a small regularizer ensuring positive definiteness and improved conditioning, and $w(d) \geq 0$ is a distance-dependent activation weight that increases as the configuration approaches inequality boundaries. Eq. (10) admits a GN interpretation, retaining the dominant term of the barrier Hessian as shown in Appendix A, while leaving the subsequent metric construction unchanged.

Since admissible motions must also satisfy equality constraints, we restrict the metric to the tangent space of the equality-constrained manifold $\mathcal{M}$. Using the equality Jacobian $J(q)$, we define the G-compatible tangent projector

$$P(q) = I - G^{-1}(q)J(q)^\top \big(J(q)G^{-1}(q)J(q)^\top\big)^{-1} J(q), \tag{11}$$

whose derivation is provided in Appendix B. The induced tangent-space metric is then given by

$$M(q) = P(q)^\top G(q) P(q). \tag{12}$$

All subsequent metric-aware operations, including steering and nearest-neighbor selection, are performed with respect to $M(q)$, and the overall construction of (G, P, M) is summarized in Algorithm 1.

C. Metric-Induced Tangent-Space Steering

Building on the Riemannian barrier metric constructed in Section III-B, we introduce a metric-induced steering rule that generates local motions on the equality-constrained manifold while respecting the geometry induced by inequality constraints. The key idea is to select, within the tangent space, a direction that minimizes the local energy measured by the induced metric, thereby biasing expansions toward admissible regions,

Algorithm 2 METRICSTEER$(q, q_{\text{target}}, \eta)$: Metric-induced tangent-space steering

Require: Feasible node $q \in \mathcal{M}$, target sample q_{target}, step size $\eta > 0$
Ensure: New configuration q_{new}

```
w ← q_target − q
(G, P, M) ← BUILDMETRIC(q)                    (Alg. 1)
w̃ ← P w                                       (tangent projection)
if ‖w̃‖ < ε then
    return q
end if
Solve M v_tmp = w̃                             (i.e., v_tmp = M^{-1} w̃)
v_0 ← P v_tmp                                 (re-project for numerical robustness)
s ← √(v_0^⊤ M v_0)
if s < ε then
    return q
end if
v̂ ← v_0 / s                                   (unit M-norm direction)
q_new ← q + η v̂                               (fixed-length step)
return q_new
```

as conceptually illustrated in Fig. 3a. Consider a tree node $q \in \mathcal{M}$ and an ambient target sample q_{target}. The ambient displacement is defined as

$$w = q_{\text{target}} - q. \tag{13}$$

Since a direct step along w generally violates the equality constraints, we first project it onto the tangent space using the metric-compatible projector $P(q)$ defined in Eq. (11):

$$\tilde{w} = P(q)\, w \in T_q\mathcal{M}. \tag{14}$$

Let $G(q) \succ 0$ denote the ambient barrier metric in Eq. (10), and let $M(q)$ denote the induced tangent-space metric defined

Algorithm 3 METRICNEAREST($q_{\text{rand}}, \mathcal{V}, K$): metric-aware nearest-neighbor selection

Require: Sample q_{rand}, tree nodes $\mathcal{V} \subset \mathcal{M}$, candidate size K
Ensure: Nearest node q_{near}

$\mathcal{V}_K \leftarrow$ EUCLIDEANKNN($q_{\text{rand}}, \mathcal{V}, K$)
$q_{\text{near}} \leftarrow \emptyset,\ d_{\min} \leftarrow +\infty$
for all $q \in \mathcal{V}_K$ **do**
  $(G, P, M) \leftarrow$ BUILDRBMETRIC(q) (Alg. 1)
  $r \leftarrow q_{\text{rand}} - q$
  $\tilde{r} \leftarrow P\,r$
  $d \leftarrow \sqrt{\tilde{r}^\top M\,\tilde{r}}$ ($\|\tilde{r}\|_{M(q)}$)
  if $d < d_{\min}$ **then**
    $d_{\min} \leftarrow d,\ \ q_{\text{near}} \leftarrow q$
  end if
end for
return q_{near}

in Eq. (12). Among tangent directions, we select the metric-induced steering direction by solving

$$v^\star = \arg\min_{v \in T_q\mathcal{M}} \ \frac{1}{2}\, v^\top M(q)\, v \quad \text{s.t.} \quad v^\top \tilde{w} = 1. \tag{15}$$

This optimization biases tangent-space steering away from inequality-boundary directions by assigning higher metric cost to boundary-approaching motions. As derived in Appendix C, Eq.(15) admits a closed-form solution, which can be interpreted as reweighting $\tilde{w}$ by the inverse induced metric, yielding a numerically robust tangent direction. The constraint $v^\top \tilde{w} = 1$ fixes the scale only for direction selection; the actual expansion magnitude is controlled separately by the step size in the update rule below. To execute a fixed-length expansion under the induced geometry, we normalize the resulting tangent direction with respect to the $M(q)$-norm and update the configuration as

$$\hat{v} = \frac{v_0}{\sqrt{v_0^\top M(q)\, v_0}}, \qquad q_{\text{new}} = q + \eta\, \hat{v}, \tag{16}$$

where $\eta > 0$ is the step size. Compared to Euclidean tangent-space steering, the proposed rule explicitly encodes inequality-induced geometry: as the configuration approaches inequality boundaries, the barrier metric increases the local energy measured by $M(q)$, thereby suppressing expansions toward infeasible regions. Consequently, the planner discourages boundary-seeking motions without relying solely on repeated collision-based rejection, leading to more stable and informative local expansions. The complete procedure is summarized in Algorithm 2.

D. Metric-Aware Sampling and Nearest-Neighbor Selection

In addition to modifying the local steering behavior, the proposed RMRRT framework incorporates the Riemannian barrier metric into sampling and nearest-neighbor selection. Random sampling is performed in the ambient configuration space, as in standard RRT-based planners, in order to preserve the simplicity and exploration properties of conventional RRT sampling. Equality consistency is subsequently enforced during the steering operation through tangent-space projection, as described in Section III-C. Given a sampled configuration q_{rand} and a set of tree nodes $\mathcal{V}$ lying on the equality-constrained manifold $\mathcal{M}$, RMRRT selects the nearest node according to a

Algorithm 4 Overall RMRRT Framework

Require: Feasible start/goal configurations $q_{\text{start}}, q_{\text{goal}} \in \mathcal{M}$; step size η; candidate size K; maximum iterations $N_{\max}$
Ensure: Feasible path σ or FAILURE

Initialize two trees $\mathcal{T}_a \leftarrow \{q_{\text{start}}\}$, $\mathcal{T}_b \leftarrow \{q_{\text{goal}}\}$
for $n = 1$ to $N_{\max}$ **do**
  $q_{\text{rand}} \sim$ UNIFORMSAMPLE($\mathbb{R}^n$)
  $q_{\text{near}} \leftarrow$ METRICNEAREST($q_{\text{rand}}, \mathcal{V}(\mathcal{T}_a), K$) (Alg. 3)
  $q_{\text{new}} \leftarrow$ METRICSTEER($q_{\text{near}}, q_{\text{rand}}, \eta$) (Alg. 2)

  $isStateValid \leftarrow$ STATEVALID(q_{new})
  $isEdgeValid \leftarrow$ EDGEVALID($q_{\text{near}}, q_{\text{new}}$)

  if not $isStateValid$ **or not** $isEdgeValid$ **then**
    continue
  end if
  Add q_{new} to $\mathcal{T}_a$ with parent q_{near}
  $q_{\text{curr}} \leftarrow$ METRICNEAREST($q_{\text{new}}, \mathcal{V}(\mathcal{T}_b), K$) (Alg. 3)
  repeat
    $q_{\text{next}} \leftarrow$ METRICSTEER($q_{\text{curr}}, q_{\text{new}}, \eta$)
    $isStateValid \leftarrow$ STATEVALID(q_{next})
    $isEdgeValid \leftarrow$ EDGEVALID($q_{\text{curr}}, q_{\text{next}}$)

    if not $isStateValid$ **or not** $isEdgeValid$ **then**
      break
    end if
    Add q_{next} to $\mathcal{T}_b$ with parent q_{curr}
    $q_{\text{curr}} \leftarrow q_{\text{next}}$

  until $\|q_{\text{curr}} - q_{\text{new}}\| < \epsilon_{\text{conn}}$
  if $\|q_{\text{curr}} - q_{\text{new}}\| < \epsilon_{\text{conn}}$ **then**
    $\sigma \leftarrow$ EXTRACTPATH($\mathcal{T}_a, q_{\text{new}}, \mathcal{T}_b, q_{\text{curr}}$)

    return σ
  end if
  Swap $(\mathcal{T}_a, \mathcal{T}_b)$
end for

return FAILURE

distance measure induced by the local Riemannian geometry. Specifically, for each node $q \in \mathcal{V}$, we define the metric-aware distance

$$d_M(q, q_{\text{rand}}) = \big\| P(q)\,(q_{\text{rand}} - q) \big\|_{M(q)} \tag{17}$$

where $P(q)$ denotes the metric-compatible tangent projection and $\|\cdot\|_{M(q)}$ is the norm induced by the Riemannian metric $M(q)$ constructed in Section III-B. In Algorithm 3, this distance is evaluated using the local metric constructed by Algorithm 1 and compared against the current minimum to identify the nearest neighbor under the constraint-induced geometry. The nearest neighbor is then selected as

$$q_{\text{near}} = \arg\min_{q \in \mathcal{V}} d_M(q, q_{\text{rand}}) \tag{18}$$

As conceptually illustrated in Fig. 3b, proximity in the ambient Euclidean configuration space does not necessarily imply feasibility on the equality-constrained manifold. Multiple tree nodes may appear close to a sampled configuration q_{rand}, while some of them lie inside infeasible regions formed

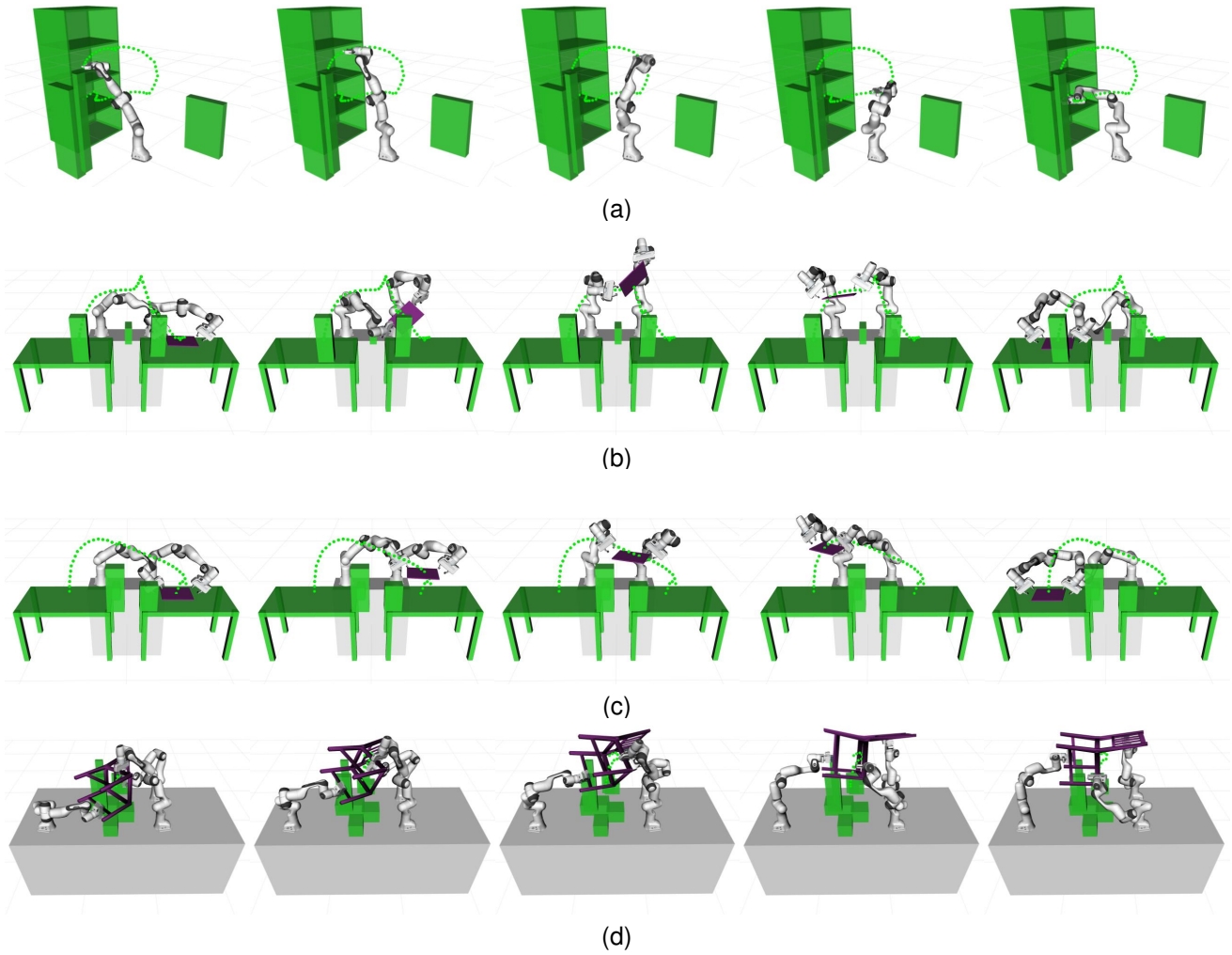

Fig. 4: Planned trajectories in four representative manipulation scenarios. The dotted curve indicates the end-effector (or object) trajectory, and snapshots depict key configurations along each path. (a) Single-arm cabinet reaching with fixed orientation constraint. (b) Dual-arm constrained transfer. (c) Dual-arm transfer with fixed orientation constraint. (d) Triple-arm constrained manipulation.

TABLE I: Planning performance across four experimental scenarios.

Planner	**Scenario 1**		**Scenario 2**		**Scenario 3**		**Scenario 4**	
	Time (s)	**Succ. (%)**	**Time (s)**	**Succ. (%)**	**Time (s)**	**Succ. (%)**	**Time (s)**	**Succ. (%)**
CBiRRT [21]	0.35 ± 0.41	100	2.97 ± 4.48	100	3.06 ± 4.27	100	74.75 ± 118.59	90
Atlas RRT* [13]	5.84 ± 12.63	100	48.55 ± 85.13	83	35.12 ± 69.27	89	10.02 ± 31.80	89
Tangent Bundle RRT* [14]	5.74 ± 16.75	100	37.71 ± 86.93	80	48.83 ± 86.27	87	15.80 ± 57.35	84
Precomputed Graph RRT [32]	2.65 ± 6.64	100	4.49 ± 5.43	100	5.51 ± 10.41	100	32.37 ± 81.47	97
LJCMP [6]	0.41 ± 1.02	100	0.24 ± 0.53	100	0.48 ± 1.56	100	0.55 ± 1.69	100
RMRRT (ours)	$\mathbf{0.32 \pm 0.34}$	100	$\mathbf{0.18 \pm 0.15}$	100	$\mathbf{0.25 \pm 0.26}$	100	$\mathbf{0.38 \pm 0.51}$	100

by inequality constraints. By evaluating distances using the Riemannian barrier metric projected onto the tangent space, RMRRT favors nodes whose local tangent spaces admit feasible, low-cost motions toward the sampled target, while naturally avoiding nodes located inside such infeasible regions. In practice, the metric-aware nearest-neighbor search is restricted to the K Euclidean-nearest candidates in the ambient space for efficiency; see the Appendix D for details.

E. Overall RMRRT Algorithm

Algorithm 4 summarizes the overall RMRRT framework. The proposed method integrates Riemannian barrier metric-based nearest-neighbor selection and tangent-space steering into a bidirectional RRT-Connect structure, thereby preserving the simplicity and scalability of standard sampling-based planning while jointly accounting for equality and inequality constraints. RMRRT draws random samples q_{rand} in the ambient configuration space and selects the nearest node q_{near} from the active tree using the metric-aware criterion

in Section III-D. It then performs a local expansion using the metric-induced tangent-space steering rule in Section III-C, which generates an equality-consistent local motion while biasing the expansion direction according to the local geometry induced by inequality constraints.

After a candidate state is generated, hard feasibility is enforced through standard state and edge validity checks. If the expansion is valid, the new node is added to the active tree, and the opposite tree is repeatedly extended toward it in a connect phase. In our implementation, the connection seed in the opposite tree is also selected using the metric-aware nearest-neighbor rule, so that both tree expansions consistently reflect the same constraint-induced geometry. The connect phase is repeated until the two trees are connected within ϵ_{conn} or the extension becomes trapped. The local Riemannian barrier metric used by the nearest-neighbor and steering modules is constructed on demand via Algorithm 1. Reusing the same local metric preserves geometric consistency between global exploration and local expansion. In addition, a lightweight projection step may be applied when numerical drift from the equality-constrained manifold is observed. By encoding inequality effects directly into the local geometry, RMRRT remains compatible with existing RRT-Connect pipelines while improving exploration robustness and expansion quality in constrained environments.

IV. Experiments

This section evaluates RMRRT under simultaneous equality and inequality constraints to assess its reliability, efficiency, and practical applicability. Simulation benchmarks are presented across diverse manipulation scenarios, including closed-chain, multi-arm, and cluttered environments. Planning performance is quantified using success rate, planning time, and path quality metrics. An ablation study is conducted in a densely cluttered dual-arm setting to isolate the contribution of the Riemannian barrier metric through comparisons between variants with and without the barrier term. Finally, real-world experiments with a panda triple-arm system for collaborative manipulation and a single humanoid platform are conducted to validate the proposed method in physical environments.

A. Simulation Benchmarks

Table I summarizes planning performance across four scenarios of increasing constraint complexity, and Fig. 4 visualizes the corresponding tasks and representative trajectories. RMRRT achieves a 100% success rate in all scenarios, indicating that the proposed metric-consistent exploration strategy reliably preserves equality constraints while expanding within regions shaped by inequality constraints. Manifold-based planners such as CBiRRT [21], Atlas-RRT [13], and Tangent Bundle RRT [13], [14] also operate on the equality-constrained manifold by constructing local tangent spaces and correcting drift via projection. However, their tangent-space machinery is typically used primarily for local expansion and is not consistently integrated into the overall exploration strategy. In particular, nearest-node selection is commonly performed using ambient Euclidean distance, which does not adequately reflect intrinsic connectability on the constraint manifold, as conceptually illustrated in Fig. 3b. By contrast, RMRRT incorporates the induced geometry into both selection and steering: nearest nodes are ranked using the Riemannian distance induced by $M(q)$ (Section III-D), and steering is performed through metric-induced reweighting under the same induced metric (Section III-C). This coupling prioritizes nodes that admit effective equality-consistent motion toward the target and suppresses boundary-seeking motions at an earlier stage, thereby reducing collision-driven rejection, shortening mean planning time, and improving success rate. In addition, the combination of geometry-aware selection and metric-based steering makes successful extensions more uniform across trials, which contributes to a lower standard deviation of planning time.

We also compare against Precomputed Graph RRT [32] and LJCMP [6]. As shown in Table I, Precomputed Graph RRT exhibits stable performance in simpler settings but degrades noticeably in Scenario 4, suggesting that a fixed offline structure may not sufficiently capture local geometric variations in high-dimensional constrained spaces. In contrast, LJCMP maintains high success rates and low average planning times across all scenarios, demonstrating the effectiveness of learning-based priors in guiding sampling toward favorable regions. RMRRT provides a competitive alternative that achieves robust performance with low variance across all scenarios without relying on learning or precomputed roadmaps. Beyond serving as a standalone planner, RMRRT can also be viewed as a geometric enhancement layer: integrating Riemannian geometry into nearest-neighbor selection and steering provides a principled mechanism for improving geometric consistency during exploration, and can be incorporated as a complementary structural module in learning-based planners to enhance both efficiency and stability.

B. Ablation Study on the Riemannian Barrier Metric

To isolate the contribution of the proposed Riemannian barrier metric from other algorithmic components, we conduct an ablation study on Unitree H1_2 single-humanoid dual-arm constrained transfer tasks. The study is performed on two scenarios with different obstacle complexities: Scenario 5 consists of a relatively simple obstacle layout, whereas Scenario 6 involves a more complex obstacle environment. This setup enables us to examine the trade-off between the additional computation introduced by the barrier-aware formulation and the exploration benefit obtained as obstacle complexity increases.

We compare (i) the full RMRRT formulation with the inequality-aware Riemannian barrier metric and (ii) a no-barrier variant in which the barrier term is disabled, so that the metric no longer captures local geometry near inequality boundaries. To ensure a fair comparison, all other components are kept identical across the two variants, including the sampling strategy, nearest-neighbor selection structure, steering pipeline, state/edge validity checks, termination criteria, and key hyperparameters. Accordingly, the observed performance differences can be primarily attributed to how effectively the barrier metric encodes inequality-induced local geometry.

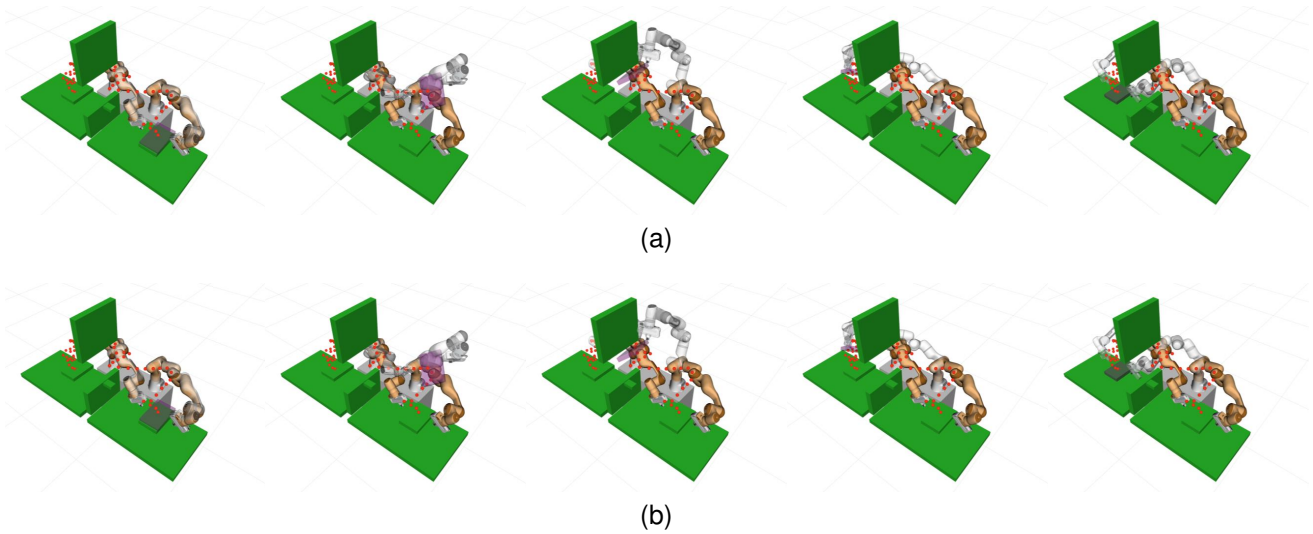

Fig. 5: Barrier-metric ablation on Unitree H1_2 single-humanoid dual-arm constrained transfer tasks with different obstacle complexities. Each panel compares representative trajectories with and without the Riemannian barrier metric: (a) Scenario 5 (simple obstacle environment) and (b) Scenario 6 (complex obstacle environment). The comparison highlights the trade-off between the additional computation of the barrier-aware formulation and the exploration benefit gained from reduced detours and more consistent expansions, with the benefit becoming more pronounced in the complex setting near inequality-induced constraint boundaries.

Fig. 5 compares representative trajectory snapshots with and without the barrier metric in both scenarios. In Scenario 5 (simple obstacle environment), both variants are generally able to complete the task successfully, while the barrier-aware variant tends to maintain more consistent expansion directions and exhibits fewer unnecessary detours. In this relatively simple setting, such differences may be qualitatively visible while remaining limited in the aggregate quantitative metrics. In contrast, in Scenario 6 (complex obstacle environment), inequality boundaries more strongly constrain locally admissible motions, and the effect of the barrier-aware metric becomes more pronounced. In particular, the barrier-enabled variant more consistently follows admissible corridors and shows fewer boundary-seeking expansions and collision-driven rejections.

Quantitative comparisons are reported in Table II (Scenario 5) and Table III (Scenario 6), using planning time, path length, rejected samples, and tangent steering failure rate as evaluation metrics. If the tangent steering failure rate remains similarly low (or unchanged) across the two variants, this indicates that the performance gap is not primarily due to improved equality-constraint satisfaction itself, but rather due to more effective steering and node selection under inequality-induced geometry. In particular, when reduced rejected samples are accompanied by lower planning time, the results suggest that the additional per-step overhead of barrier-metric evaluation is outweighed by reductions in repeated collision checks and unproductive expansions.

Overall, the Riemannian barrier metric exhibits a scenario-dependent trade-off. In simpler environments, the relative impact of the additional metric computation may be more noticeable. However, as obstacle complexity increases and inequality constraints dominate the local exploration geometry, the barrier-aware formulation reduces unnecessary detours and collision-driven rejection, thereby improving overall exploration efficiency and stability. These results support the central premise of the proposed method: its main advantage is not merely maintaining equality-constraint feasibility, but structurally incorporating inequality-induced local geometry into exploration decisions.

V. Conclusion

In this paper, we proposed RMRRT, a sampling-based motion planning framework that enables inequality-aware steering while preserving feasibility on equality-constrained manifolds. The central idea of the proposed approach is to define a Riemannian metric on the tangent space of the equality manifold that explicitly captures the geometric structure induced by inequality constraints. This metric is constructed by combining a null-space projection of the equality constraint Jacobian with the Hessian of a logarithmic barrier function associated with inequality constraints. Steering directions are then selected to minimize the induced Riemannian norm within the equality-consistent tangent space, naturally guiding tree expansion away from infeasible regions without relying on repeated projection or collision-based sample rejection. The proposed RMRRT framework was validated through a wide range of simulation benchmarks, including fixed-orientation, multi-arm, and humanoid manipulation tasks, as well as real-world experiments using dual-arm and humanoid robotic platforms. Across these evaluations, RMRRT consistently demonstrated improved planning efficiency and reliability compared to existing constrained and learning-based planners.

Future work will explore the integration of the proposed RMRRT framework with learning-based motion planning approaches. Since the proposed steering mechanism is geometrically defined on equality-constrained manifolds and operates

TABLE II: Ablation study on the effect of the Riemannian barrier metric (Scenario 5).

Method	Time (s)	Path Length (m)	Reject Samples (N)	Tangent Steering Failure (%)
RMRRT (w/o barrier)	1.12 ± 1.12	19.07 ± 1.12	383 ± 1.12	0 ± 0
RMRRT (full)	$\mathbf{0.83 \pm 1.12}$	$\mathbf{17.49 \pm 1.12}$	$\mathbf{226 \pm 1.12}$	$\mathbf{0 \pm 0}$

TABLE III: Ablation study on the effect of the Riemannian barrier metric (Scenario 6).

Method	Time (s)	Path Length (m)	Reject Samples (N)	Tangent Steering Failure (%)
RMRRT (w/o barrier)	46.48 ± 19.30	31.74 ± 13.49	693.46 ± 312.35	0 ± 0
RMRRT (full)	$\mathbf{0.83 \pm 1.12}$	$\mathbf{17.49 \pm 1.12}$	$\mathbf{226 \pm 1.12}$	$\mathbf{0 \pm 0}$

independently of assumptions on sampling distributions or latent representations, it has strong potential to complement learning-based planners by enhancing local expansion efficiency when combined with learned sampling strategies, heuristics, or latent-space exploration.

Appendix A
Gauss–Newton Approximation of the Barrier Metric

This appendix justifies the Jacobian-based ambient metric in Eq. (10) as a Gauss–Newton (GN) surrogate of the ideal Hessian accumulation in Eq. (9). For a distance-based inequality constraint with signed distance $d(q)$ and barrier $\phi(q) = \phi(d(q))$, the chain rule gives

$$\nabla^2\phi(q) = \phi''(d)\,\nabla d(q)\nabla d(q)^\top \;+\; \phi'(d)\,\nabla^2 d(q). \tag{A1}$$

Near closest-feature changes and contact switching, $\nabla^2 d(q)$ can be ill-conditioned or undefined; thus we adopt the GN approximation

$$\nabla^2\phi(q) \;\approx\; \phi''(d)\,\nabla d(q)\nabla d(q)^\top, \tag{A2}$$

which retains the dominant PSD term. Using a geometric proxy point $p(q) \in \mathbb{R}^3$ with Jacobian $J(q) = \partial p(q)/\partial q \in \mathbb{R}^{3\times n}$ and distance normal $n(q) = \partial d/\partial p \in \mathbb{R}^3$, we have

$$\nabla d(q) \;=\; n(q)^\top J(q), \tag{A3}$$

and substituting (A3) into (A2) yields

$$\nabla^2\phi(q) \;\approx\; \phi''(d)\,J(q)^\top\big(n(q)n(q)^\top\big)J(q) \;\preceq\; \phi''(d)\,J(q)^\top J(q), \tag{A4}$$

since $n(q)n(q)^\top \preceq I$. Absorbing $\phi''(d)$ into a nonnegative weight $w(d) \geq 0$ and summing over m proxy points gives the practical GN barrier metric

$$G(q) \;=\; \lambda I \;+\; \sum_{j=1}^{m} w(d_j(q))\; J_j(q)^\top J_j(q), \tag{A5}$$

where $\lambda > 0$ ensures positive definiteness and improves conditioning.

Appendix B
Derivation of the Metric-Aware Projection Operator

Consider the equality-constrained manifold $\mathcal{M} = \{q \in \mathbb{R}^n \mid g(q) = 0\}$ with Jacobian $J(q) = \partial g/\partial q \in \mathbb{R}^{m\times n}$ and tangent space $T_q\mathcal{M} = \{v \in \mathbb{R}^n \mid J(q)v = 0\}$. Given $w \in \mathbb{R}^n$ and an SPD metric $G(q)$, the G-orthogonal projection of w onto $T_q\mathcal{M}$ is defined by

$$\min_{v\in\mathbb{R}^n}\; \frac{1}{2}(v-w)^\top G(q)(v-w) \quad \text{s.t.} \quad J(q)v = 0. \tag{B1}$$

The KKT conditions are

$$G(q)(v-w) + J(q)^\top\lambda = 0, \qquad J(q)v = 0. \tag{B2}$$

Eliminating v gives

$$\lambda = \big(J(q)G^{-1}(q)J(q)^\top\big)^{-1}J(q)w, \tag{B3}$$

and hence $v = P(q)w$ with

$$P(q) = I - G^{-1}(q)J(q)^\top\big(J(q)G^{-1}(q)J(q)^\top\big)^{-1}J(q). \tag{B4}$$

By construction, $J(q)P(q) = 0$, thus $P(q)w \in T_q\mathcal{M}$.

Appendix C
Derivation of the Metric-Induced Steering Direction

Let $w \in \mathbb{R}^n$ be an ambient displacement (e.g., $w = q_{\text{rand}} - q$) and let $\tilde{w} = P(q)w \in T_q\mathcal{M}$. Given a Riemannian metric $G(q) \succ 0$, define the induced tangent-space metric

$$M(q) = P(q)^\top G(q)P(q). \tag{C0}$$

To obtain a metric-induced tangent direction that minimizes local energy while making progress along $\tilde{w}$, consider

$$v^* = \arg\min_{v\in T_q\mathcal{M}}\; \frac{1}{2}\,v^\top M(q)v \quad \text{s.t.} \quad v^\top\tilde{w} = 1 \tag{C1}$$

Using $\mathcal{L}(v,\lambda) = v^\top Mv + \lambda(1 - v^\top\tilde{w})$ gives

$$v^* = \frac{M^{-1}(q)\tilde{w}}{\tilde{w}^\top M^{-1}(q)\tilde{w}} \tag{C2}$$

In implementation, we compute $v_{\text{tmp}} = M^{-1}(q)\tilde{w}$, apply the projector once more for numerical robustness,

$$v_0 = P(q)v_{\text{tmp}} = P(q)M^{-1}(q)\tilde{w} = P(q)M^{-1}(q)P(q)w \tag{C3}$$

and take a fixed-length step using the unit $M(q)$-norm direction

$$\hat{v} = \frac{v_0}{\sqrt{v_0^\top M(q)\,v_0}}, \qquad q^+ = q + \eta\,\hat{v}, \tag{C4}$$

where $\eta > 0$ is the step size.

APPENDIX D
METRIC-AWARE NEAREST-NEIGHBOR EVALUATION

Given a sample $q_{\text{rand}} \in \mathbb{R}^n$ and tree nodes $\mathcal{V} \subset \mathcal{M}$, the metric-aware nearest neighbor is

$$q_{\text{near}} = \arg\min_{q \in \mathcal{V}} \left\| P(q)(q_{\text{rand}} - q) \right\|_{M(q)}. \tag{D1}$$

In practice, we restrict evaluation to the K Euclidean-nearest candidates in the ambient space,

$$\mathcal{V}_K = \arg \min_{\substack{\mathcal{S} \subset \mathcal{V} \\ |\mathcal{S}| = K}} \sum_{q \in \mathcal{S}} \|q_{\text{rand}} - q\|_2, \tag{D2}$$

and compute the metric-aware distance only over $\mathcal{V}_K$:

$$q_{\text{near}} = \arg \min_{q \in \mathcal{V}_K} \left\| P(q)(q_{\text{rand}} - q) \right\|_{M(q)}. \tag{D3}$$

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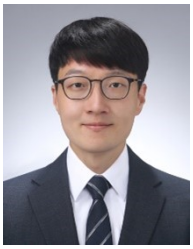

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